CHE 565 -- Homework 3

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Problem 1

Steepest Descent

Function :

$$f(x_1, x_2) = 0.5x_1^2 + 1.0x_1x_2 - x_1 + 1.0x_2^2 - x_2$$

At x0 =Matrix([[1], [1]]), f(x0) = 0.500000000000000

Gradient of f:

$$\nabla f = \left[\begin{pmatrix} 1.0x_1 + 1.0x_2 - 1.0 \\ 1.0x_1 + 2.0x_2 - 1 \end{pmatrix} \right]$$

At x0 =Matrix([[1], [1]]), Df(x0) = 1.00, 2.00

$$\begin{bmatrix} g_1(\alpha) \\ g_2(\alpha) \end{bmatrix} = \begin{bmatrix} 1 - 1.0\alpha \\ 1 - 2.0\alpha \end{bmatrix}$$

Function phi_k:

$$\phi(\alpha) = 6.5\alpha^2 - 5.0\alpha + 0.5$$

Derivative of phi_k:

$$\frac{d}{d\alpha}\phi(\alpha) = 13.0\alpha - 5.0$$
$$0 = 13.0\alpha - 5.0$$

Optimal alpha: 0.38

Next iterate x1: 0.62, x2: 0.23

Function value at next iterate: -0.46

Problem 2

Conjugate Gradient Method

$$\begin{bmatrix} s_1(\alpha) \\ s_2(\alpha) \end{bmatrix} = \begin{bmatrix} 0.0946745562130178 \\ -0.195266272189349 \end{bmatrix}$$

Problem 3

Constituents	Maximum Quantity (bbl/day)	Production Cost ($/bbl)
1	3000	26.00
2	2000	30.60
3	4000	29.20
4	1000	29.80

Grade	Specifications	Selling Price ($/bbl)
A	No more than 15% of 1; No more than 40% of 2	32.40
B	No more than 50% of 3; No more than 10% of 1	31.50
C	No less than 10% of 2; No more than 20% of 1	30.60

$$A = a_1 + a_2 + a_3 + a_4$$
$$B = b_1 + b_2 + b_3 + b_4$$
$$C = c_1 + c_2 + c_3 + c_4$$

$$a_1 \leq 0.15A$$
$$a_2 \geq 0.40A$$

$$b_3 \leq 0.50B$$
$$b_1 \leq 0.10B$$

$$c_2 \geq 0.10C$$
$$c_1 \leq 0.20C$$

$$a_1 + b_1 + c_1 \leq 3000$$
$$a_2 + b_2 + c_2 \leq 2000$$
$$a_3 + b_3 + c_3 \leq 4000$$
$$a_4 + b_4 + c_4 \leq 1000$$
$$a_i, b_i, c_i \geq 0, \quad i = 1, \ldots, 4$$

$$P = 32.40A + 31.50B + 30.60C - \left[26(a_1 + b_1 + c_1) + 30.60(a_2 + b_2 + c_2) + 29.20(a_3 + b_3 + c_3) + 29.80(a_4 + b_4 + c_4)\right]$$

$$P = 32.4A + 31.5B + 30.6C - 26.0a_1 - 30.6a_2 - 29.2a_3 - 29.8a_4 - 26.0b_1 - 30.6b_2 - 29.2b_3 - 29.8b_4 - 26.0c_1 - 30.6c_2 - 29.2c_3 - 29.8c_4$$